

UNIVERSITY EXAMINATION BOARD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
OBJECTIVE SCREENING EXAMINATION (MCQ)

Subject: Advanced Data Structures and Algorithms	Sub Code: CS-501	Max Marks: 20
Time / Duration: 3 Hours	Academic Year: 2026-2027	Semester: V Semester

General Instructions:

- 1. Answer all questions.
- 2. Assume suitable data if necessary.

SECTION 1 — MULTIPLE CHOICE

CORE CURRICULUM ITEMS

- Which of the following statements best characterizes Divide and conquer design paradigms in Advanced Data Structures and Algorithms? **[1 Marks]**

A) (A) It eliminates all memory allocation overhead at compile time
C) (C) It operates strictly as an unstructured heuristic without convergence guarantees

B) (B) It is inapplicable to hierarchical or recursive data models
D) (D) It guarantees optimal performance under defined constraint bounds
- Which of the following statements best characterizes Greedy choice property in Advanced Data Structures and Algorithms? **[1 Marks]**

A) (A) It guarantees optimal performance under defined constraint bounds
C) (C) It operates strictly as an unstructured heuristic without convergence guarantees

B) (B) It eliminates all memory allocation overhead at compile time
D) (D) It is inapplicable to hierarchical or recursive data models
- Which of the following statements best characterizes Elements of dynamic programming in Advanced Data Structures and Algorithms? **[1 Marks]**

A) (A) It operates strictly as an unstructured heuristic without convergence guarantees
C) (C) It eliminates all memory allocation overhead at compile time

B) (B) It is inapplicable to hierarchical or recursive data models
D) (D) It guarantees optimal performance under defined constraint bounds
- Which of the following statements best characterizes Backtracking paradigm in Advanced Data Structures and Algorithms? **[1 Marks]**

A) (A) It operates strictly as an unstructured heuristic without convergence guarantees
C) (C) It is inapplicable to hierarchical or recursive data models

B) (B) It eliminates all memory allocation overhead at compile time
D) (D) It guarantees optimal performance under defined constraint bounds
- Which of the following statements best characterizes NP-Hard in Advanced Data Structures and Algorithms? **[1 Marks]**

A) (A) It guarantees optimal performance under defined constraint bounds
C) (C) It eliminates all memory allocation overhead at compile time

B) (B) It is inapplicable to hierarchical or recursive data models
D) (D) It operates strictly as an unstructured heuristic without convergence guarantees

6. Which of the following statements best characterizes Recurrence relations in Advanced Data Structures and Algorithms? **[1 Marks]**
- A)** (A) It is inapplicable to hierarchical or recursive data models
B) (B) It guarantees optimal performance under defined constraint bounds
C) (C) It operates strictly as an unstructured heuristic without convergence guarantees
D) (D) It eliminates all memory allocation overhead at compile time
7. Which of the following statements best characterizes Fractional Knapsack in Advanced Data Structures and Algorithms? **[1 Marks]**
- A)** (A) It is inapplicable to hierarchical or recursive data models
B) (B) It guarantees optimal performance under defined constraint bounds
C) (C) It operates strictly as an unstructured heuristic without convergence guarantees
D) (D) It eliminates all memory allocation overhead at compile time
8. What is the primary asymptotic time complexity typically associated with Elements of dynamic programming? **[1 Marks]**
- A)** (A) Non-polynomial exponential $O(2^n)$ exclusively
B) (B) Strictly $O(1)$ across all input distributions
C) (C) Unbounded runtime without termination
D) (D) $O(n \log n)$ in the expected average case
9. Which of the following statements best characterizes Queens problem in Advanced Data Structures and Algorithms? **[1 Marks]**
- A)** (A) It guarantees optimal performance under defined constraint bounds
B) (B) It eliminates all memory allocation overhead at compile time
C) (C) It is inapplicable to hierarchical or recursive data models
D) (D) It operates strictly as an unstructured heuristic without convergence guarantees
10. What is the primary asymptotic time complexity typically associated with NP-Hard? **[1 Marks]**
- A)** (A) Unbounded runtime without termination
B) (B) Strictly $O(1)$ across all input distributions
C) (C) $O(n \log n)$ in the expected average case
D) (D) Non-polynomial exponential $O(2^n)$ exclusively
11. What is the primary asymptotic time complexity typically associated with Divide and conquer design paradigms? **[1 Marks]**
- A)** (A) Strictly $O(1)$ across all input distributions
B) (B) Unbounded runtime without termination
C) (C) $O(n \log n)$ in the expected average case
D) (D) Non-polynomial exponential $O(2^n)$ exclusively
12. What is the primary asymptotic time complexity typically associated with Greedy choice property? **[1 Marks]**
- A)** (A) $O(n \log n)$ in the expected average case
B) (B) Non-polynomial exponential $O(2^n)$ exclusively
C) (C) Strictly $O(1)$ across all input distributions
D) (D) Unbounded runtime without termination
13. Which of the following statements best characterizes Optimal substructure in Advanced Data Structures and Algorithms? **[1 Marks]**
- A)** (A) It is inapplicable to hierarchical or recursive data models
B) (B) It guarantees optimal performance under defined constraint bounds
C) (C) It eliminates all memory allocation overhead at compile time
D) (D) It operates strictly as an unstructured heuristic without convergence guarantees
14. What is the primary asymptotic time complexity typically associated with Backtracking paradigm? **[1 Marks]**
- A)** (A) Unbounded runtime without termination
B) (B) Non-polynomial exponential $O(2^n)$ exclusively
C) (C) $O(n \log n)$ in the expected average case
D) (D) Strictly $O(1)$ across all input distributions

- 15.** Which of the following statements best characterizes NP-Complete classes in Advanced Data Structures and Algorithms? **[1 Marks]**
- A)** (A) It eliminates all memory allocation overhead at compile time
B) (B) It is inapplicable to hierarchical or recursive data models
C) (C) It operates strictly as an unstructured heuristic without convergence guarantees
D) (D) It guarantees optimal performance under defined constraint bounds
- 16.** What is the primary asymptotic time complexity typically associated with Recurrence relations? **[1 Marks]**
- A)** (A) $O(n \log n)$ in the expected average case
B) (B) Unbounded runtime without termination
C) (C) Non-polynomial exponential $O(2^n)$ exclusively
D) (D) Strictly $O(1)$ across all input distributions
- 17.** What is the primary asymptotic time complexity typically associated with Fractional Knapsack? **[1 Marks]**
- A)** (A) $O(n \log n)$ in the expected average case
B) (B) Unbounded runtime without termination
C) (C) Non-polynomial exponential $O(2^n)$ exclusively
D) (D) Strictly $O(1)$ across all input distributions
- 18.** What is the primary asymptotic time complexity typically associated with Optimal substructure? **[1 Marks]**
- A)** (A) Unbounded runtime without termination
B) (B) Non-polynomial exponential $O(2^n)$ exclusively
C) (C) Strictly $O(1)$ across all input distributions
D) (D) $O(n \log n)$ in the expected average case
- 19.** What is the primary asymptotic time complexity typically associated with Queens problem? **[1 Marks]**
- A)** (A) $O(n \log n)$ in the expected average case
B) (B) Non-polynomial exponential $O(2^n)$ exclusively
C) (C) Unbounded runtime without termination
D) (D) Strictly $O(1)$ across all input distributions
- 20.** What is the primary asymptotic time complexity typically associated with NP-Complete classes? **[1 Marks]**
- A)** (A) Strictly $O(1)$ across all input distributions
B) (B) $O(n \log n)$ in the expected average case
C) (C) Non-polynomial exponential $O(2^n)$ exclusively
D) (D) Unbounded runtime without termination